

Answer: C

16 For simple harmonic motion,

$$E_k = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2(x_0^2 - x^2)$$

Thus the graph for E_k should be an inverted parabolic curve.

E_k is maximum at $x = 0$.

$$E_T = E_{k,\max} = \frac{1}{2}m\omega^2x_0^2$$

$$E_P = E_T - E_k = \frac{1}{2}m\omega^2x_0^2 - \frac{1}{2}m\omega^2(x_0^2 - x^2) = \frac{1}{2}m\omega^2x^2$$

Thus the graph for E_P should be an upright parabolic curve.

Answer: A

17 Since the simple pendulum undergoes simple harmonic motion